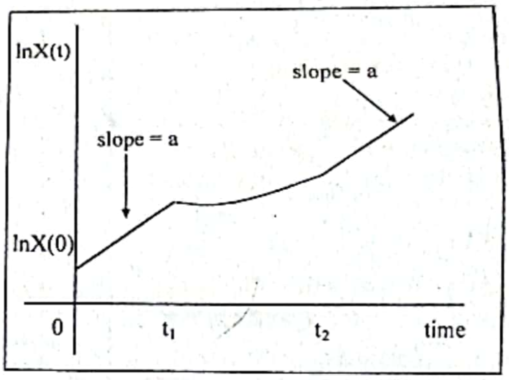
(b) Note that the slope of $\ln X(t)$ plotted against time is equal to the growth rate of $X(t)$. That is, we know

$$\frac{d \ln X(t)}{dt} = \frac{\dot{X}(t)}{X(t)}$$

(See equation (1.10) in the text.)

From time 0 to time t_1 the slope of $\ln X(t)$ equals $a > 0$. The $\ln X(t)$ locus has an inflection point at t_1, when the growth rate of $X(t)$ changes discontinuously from a to 0. Between t_1 and t_2, the slope of $\ln X(t)$ rises gradually from 0 to a. After time t_2 the slope of $\ln X(t)$ is constant and equal to $a > 0$ again.

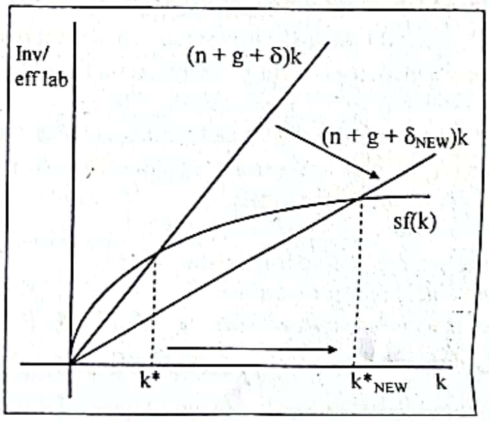


Problem 1.3

(a) The slope of the break-even investment line is given by $(n + g + \delta)$ and thus a fall in the rate of depreciation, δ, decreases the slope of the break-even investment line.

The actual investment curve, $sf(k)$ is unaffected.

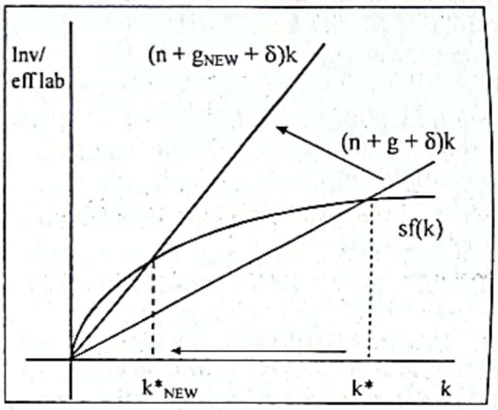
From the figure at right we can see that the balanced-growth-path level of capital per unit of effective labor rises from k^* to k^*_{NEW}.



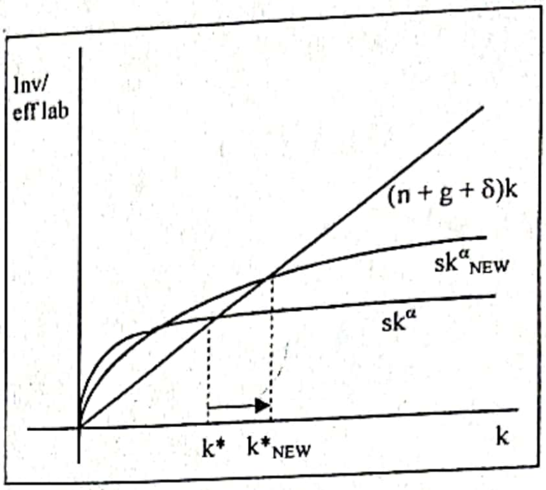
(b) Since the slope of the break-even investment line is given by $(n + g + \delta)$, a rise in the rate of technological progress, g, makes the break-even investment line steeper.

The actual investment curve, $sf(k)$, is unaffected.

From the figure at right we can see that the balanced-growth-path level of capital per unit of effective labor falls from k^* to k^*_{NEW}.



(c) The break-even investment line, $(n + g + \delta)k$, is unaffected by the rise in capital's share, α.



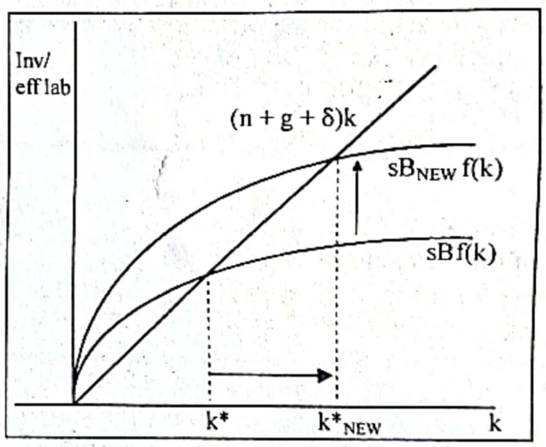
The effect of a change in α on the actual investment curve, sk^{α}, can be determined by examining the derivative $\partial(sk^{\alpha})/\partial\alpha$. It is possible to show that

(1) $\dfrac{\partial sk^{\alpha}}{\partial \alpha} = sk^{\alpha} \ln k$.

For $0 < \alpha < 1$, and for positive values of k, the sign of $\partial(sk^{\alpha})/\partial\alpha$ is determined by the sign of lnk. For $\ln k > 0$, or $k > 1$, $\partial sk^{\alpha}/\partial\alpha > 0$ and so the new actual investment curve lies above the old one. For $\ln k < 0$ or $k < 1$, $\partial sk^{\alpha}/\partial\alpha < 0$ and so the new actual investment curve lies below the old one. At $k = 1$, so that $\ln k = 0$, the new actual investment curve intersects the old one.

In addition, the effect of a rise in α on k^* is ambiguous and depends on the relative magnitudes of s and $(n + g + \delta)$. It is possible to show that a rise in capital's share, α, will cause k^* to rise if $s > (n + g + \delta)$. This is the case depicted in the figure above.

(d) Suppose we modify the intensive form of the production function to include a non-negative constant, B, so that the actual investment curve is given by $sBf(k)$, $B > 0$.



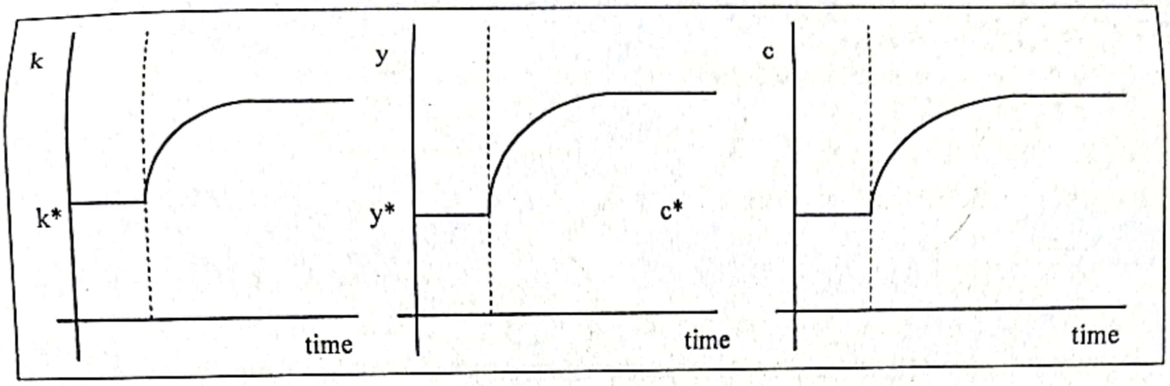
Then workers exerting more effort, so that output per unit of effective labor is higher than before, can be modeled as an increase in B. This increase in B shifts the actual investment curve up.

The break-even investment line, $(n + g + \delta)k$, is unaffected.

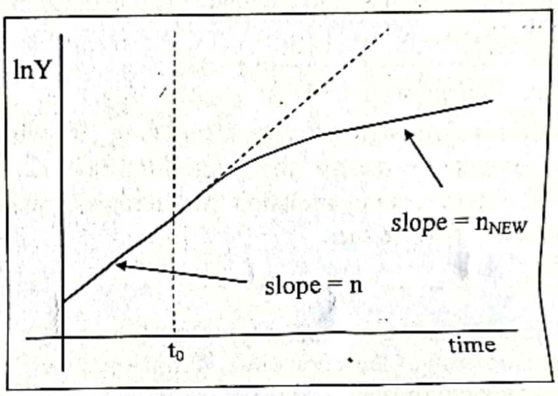
From the figure at right we can see that the balanced-growth-path level of capital per unit of effective labor rises from k^* to k^*_{NEW}.

Problem 1.4

(a) At some time, call it t_0, there is a discrete upward jump in the number of workers. This reduces the amount of capital per unit of effective labor from k^* to k_{NEW}. We can see this by simply looking at the definition, $k \equiv K/AL$. An increase in L without a jump in K or A causes k to fall. Since $f'(k) > 0$, this fall in the amount of capital per unit of effective labor reduces the amount of output per unit of effective labor as well. In the figure below, y falls from y^* to y_{NEW}.



(b) By definition, output can be written as $Y \equiv Ly$. Thus the growth rate of output is $\dot{Y}/Y = \dot{L}/L + \dot{y}/y$. On the initial balanced growth path, $\dot{y}/y = 0$ – output per worker is constant – so $\dot{Y}/Y = \dot{L}/L = n$. On the final balanced growth path, $\dot{y}/y = 0$ again – output per worker is constant again – and so $\dot{Y}/Y = \dot{L}/L = n_{NEW} < n$. In the end, output will be growing at a *permanently* lower rate.



What happens during the transition? Examine the production function Y = F(K,AL). On the initial balanced growth path AL, K and thus Y are all growing at rate n. Then suddenly AL begins growing at some new lower rate n_{NEW}. Thus suddenly Y will be growing at some rate between that of K (which is growing at n) and that of AL (which is growing at n_{NEW}). Thus, during the transition, output grows more rapidly than it will on the new balanced growth path, but less rapidly than it would have without the decrease in population growth. As output growth gradually slows down during the transition, so does capital growth until finally K, AL, and thus Y are all growing at the new lower n_{NEW}.

Problem 1.7

The derivative of $y^* = f(k^*)$ with respect to n is given by

(1) $\partial y^*/\partial n = f'(k^*)[\partial k^*/\partial n]$.

To find $\partial k^*/\partial n$, use the equation for the evolution of the capital stock per unit of effective labor, $\dot{k} = sf(k) - (n+g+\delta)k$. In addition, use the fact that on a balanced growth path, $\dot{k} = 0$, $k = k^*$ and thus $sf(k^*) = (n + g + \delta)k^*$. Taking the derivative of both sides of this expression with respect to n yields

(2) $$sf'(k^*)\frac{\partial k^*}{\partial n} = (n+g+\delta)\frac{\partial k^*}{\partial n} + k^*,$$

and rearranging yields

(3) $$\frac{\partial k^*}{\partial n} = \frac{k^*}{sf'(k^*) - (n+g+\delta)}.$$

Substituting equation (3) into equation (1) gives us

(4) $$\frac{\partial y^*}{\partial n} = f'(k^*)\left[\frac{k^*}{sf'(k^*) - (n+g+\delta)}\right].$$

Rearranging the condition that implicitly defines k*, sf(k*) = (n + g + δ)k*, and solving for s yields

(5) $s = (n + g + \delta)k^*/f(k^*)$.

Substitute equation (5) into equation (4):

$$(6)\quad \frac{\partial y^*}{\partial n} = \frac{f'(k^*)k^*}{[(n+g+\delta)f'(k^*)k^*/f(k^*)] - (n+g+\delta)}.$$

To turn this into the elasticity that we want, multiply both sides of equation (6) by n/y*:

$$(7)\quad \frac{n}{y^*}\frac{\partial y^*}{\partial n} = \frac{n}{(n+g+\delta)}\frac{f'(k^*)k^*/f(k^*)}{[f'(k^*)k^*/f(k^*)]-1}.$$

Using the definition that $\alpha_K(k^*) \equiv f'(k^*)k^*/f(k^*)$ gives us

$$(8)\quad \frac{n}{y^*}\frac{\partial y^*}{\partial n} = -\frac{n}{(n+g+\delta)}\left[\frac{\alpha_K(k^*)}{1-\alpha_K(k^*)}\right].$$

Now, with $\alpha_K(k^*) = 1/3$, g = 2% and δ = 3%, we need to calculate the effect on y* of a fall in n from 2% to 1%. Using the midpoint of n = 0.015 to calculate the elasticity gives us

$$(9)\quad \frac{n}{y^*}\frac{\partial y^*}{\partial n} = -\frac{0.015}{(0.015+0.02+0.03)}\left(\frac{1/3}{1-1/3}\right) \cong -0.12.$$

So this 50% drop in the population growth rate, from 2% to 1%, will lead to approximately a 6% increase in the level of output per unit of effective labor, since (-0.50)(-0.12) = 0.06. This calculation illustrates the point that observed differences in population growth rates across countries are not nearly enough to account for differences in y that we see.

Problem 1.8

(a) A permanent increase in the fraction of output that is devoted to investment from 0.15 to 0.18 represents a 20 percent increase in the saving rate. From equation (1.27) in the text, the elasticity of output with respect to the saving rate is

$$(1)\quad \frac{s}{y^*}\frac{\partial y^*}{\partial s} = \frac{\alpha_K(k^*)}{1-\alpha_K(k^*)},$$

where $\alpha_K(k^*)$ is the share of income paid to capital (assuming that capital is paid its marginal product).

Substituting the assumption that $\alpha_K(k^*) = 1/3$ into equation (1) gives us

$$(2)\quad \frac{s}{y^*}\frac{\partial y^*}{\partial s} = \frac{\alpha_K(k^*)}{1-\alpha_K(k^*)} = \frac{1/3}{1-1/3} = \frac{1}{2}.$$

Thus the elasticity of output with respect to the saving rate is 1/2. So this 20 percent increase in the saving rate – from s = 0.15 to s_{NEW} = 0.18 – causes output to rise relative to what it would have been by about 10 percent. [Note that the analysis has been carried out in terms of output per unit of effective labor. Since the paths of A and L are not affected, however, if output per unit of effective labor rises by 10 percent, output itself is also 10 percent higher than what it would have been.]

(b) Consumption rises less than output. Output ends up 10 percent higher than what it would have been. But the fact that the saving rate is higher means that we are now consuming a smaller fraction of output. We can calculate the elasticity of consumption with respect to the saving rate. On the balanced growth path, consumption is given by

(3) $c^* = (1 - s)y^*$.

Taking the derivative with respect to s yields

1

$$\left(\frac{1}{\gamma-1}\right)(P_i - P) = y + z_i - \eta(P_i - P).$$

$$\Rightarrow \frac{1}{\gamma-1} P_i + \eta P_i = y + z_i + \eta P + \frac{1}{\gamma-1} P.$$

$$\Rightarrow P_i\left(\frac{1+\eta(\gamma-1)}{\gamma-1}\right) = y + z_i + P\left(\frac{\eta(\gamma-1)+1}{\gamma-1}\right)$$

$$\Rightarrow P_i\left(\frac{1+\eta\gamma-\eta}{\gamma-1}\right) = y + z_i + P\left(\frac{\eta\gamma-\eta+1}{\gamma-1}\right)$$

$$\Rightarrow P_i = \left(\frac{\gamma-1}{1+\eta\gamma-\eta}\right)(y+z_i) + P \quad \text{———} \quad (15.22)$$

$z_i = 0$. Then,

$$P = \frac{\gamma-1}{1+\eta\gamma-\eta}\, y + P$$

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* Solving the model

$$y = b(p - E(p)) = m - p.$$

$$\Rightarrow bp + p = m + bE(p)$$

$$\Rightarrow p(1+b) = m + bE(p).$$

$$\Rightarrow p = \frac{m}{1+b} + \frac{b}{1+b} E(p) \quad \text{———} \quad (15.31).$$

$$\therefore y = m - p = m - \frac{m}{1+b} - \frac{b}{1+b} E(p)$$

$$= m\left(\frac{1+b-1}{1+b}\right) - \frac{b}{1+b} E(p).$$

$$= \frac{mb}{1+b} - \frac{b}{1+b} E(p) \quad \text{———} \quad (15.32).$$

From 15.31,

$$E(p) - \frac{b}{1+b}E(p) = \frac{1}{1+b}E(m)$$

$$\Rightarrow E(p)\left(\frac{1+b-b}{1+b}\right) = \frac{1}{1+b}E(m).$$

$$\Rightarrow E(p) = E(m) \quad \text{———} \quad (15.34)$$

From 15.31,

$$p = \frac{1}{1+b}\left(E(m)+m-E(m)\right) + \frac{b}{1+b}E(m)$$

$$[\because m = E(m)+m-E(m)]$$

$$= \frac{1}{1+b}E(m) + \frac{b}{1+b}E(m) + \frac{1}{1+b}\{m-E(m)\}$$

$$= \frac{1+b}{1+b}E(m) + \frac{1}{1+b}\{m-E(m)\}.$$

$$= E(m) + \frac{1}{1+b}\{m-E(m)\} \quad \text{———} \quad (15.35)$$

$$\therefore y = \frac{b}{1+b}\left(m - E(p)\right)$$

$$= \frac{b}{1+b}\left(m - E(m)\right) \quad [\because E(p) = E(m)]$$

$$\text{———} \quad (15.36)$$

(2)

$$\frac{\dot{Y}_t}{Y_t} = \sigma(i_t - \pi_t - \rho).$$ (3)

$$\therefore \frac{\dot{X}_t}{X_t} = \sigma(i_t - \pi_t - \rho) - g.$$

$$\therefore \dot{x}_t = \sigma(i_t - \pi_t) - \sigma\rho - g. \left[\because \frac{\dot{X}_t}{X_t} = \dot{x}_t\right]$$

$$= \sigma(i_t - \pi_t) - \sigma(\rho + \sigma^{-1} g).$$

$$= \sigma(i_t - \pi_t) - \sigma r^n \left[\because r^n = \rho + \sigma^{-1} g\right]$$

$$= \sigma(i_t - \pi_t - r^n). \quad (15.42)$$

Leibniz rule

$$g(x) = \int_{a(x)}^{b(x)} f(x,s)\, ds.$$

$$\frac{dg}{dx} = f(x, b(x))\frac{db}{dx} - f(x, a(x))\frac{da}{dx} + \int_{a(x)}^{b(x)} \frac{df(x,s)}{dx}\, ds.$$

Using Leibniz rule

$$-\dot{P}_t = -\alpha(v_t - P_t)$$

$$\Rightarrow \dot{P}_t = \alpha(v_t - P_t).$$

$$\& \quad \dot{v}_t - \dot{P}_t = -\alpha\eta x_t + \rho(v_t - P_t)$$

$$= -\alpha\eta x_t + \frac{\rho}{\alpha}\pi_t \left[\because \pi_t = \alpha(v_t - P_t)\right]$$

$$\therefore \dot{\pi}_t = \alpha(\dot{v}_t - \dot{p}_t)$$

$$= \alpha\left(-\alpha\eta x_t + \frac{\rho}{\alpha}\pi_t\right).$$

$$= -\alpha^2\eta x_t + \rho\pi_t.$$

$$= -kx_t + \rho\pi_t.$$

$$\therefore \dot{\pi}_t = \rho\pi_t - kx_t. \quad \text{———} \quad (15.52)$$

From 15.42,

$$\dot{x}_t = \sigma(i_t - \pi_t - r^n).$$

$$\therefore \bar{\pi} = i - r^n \; [\because \dot{x}_t = 0].$$

Demand

From 15.52,

$$\dot{\pi}_t = \rho\pi_t - kx_t.$$

$$\therefore \rho\pi_t = kx_t \; [\because \dot{\pi}_t = 0].$$

Supply

$$\begin{bmatrix} \dot{\pi}_t \\ \dot{x}_t \end{bmatrix} = -\Omega \begin{bmatrix} \pi_t \\ x_t \end{bmatrix} + \begin{bmatrix} 0 \\ \sigma(i - r^n) \end{bmatrix} \rightarrow [\because \dot{x}_t = \sigma(i_t - \pi_t - r^n)]$$

$$\Omega = \begin{bmatrix} \rho & -k \\ -\sigma & 0 \end{bmatrix} \begin{matrix} \rightarrow \dot{\pi}_t = \rho\pi_t - kx_t \\ \rightarrow \dot{x}_t = \sigma(i_t - \pi_t - r^n) \end{matrix}$$

$\Rightarrow \sigma(i - r^n) [\because \dot{\pi}_t = 0]$

So $-\sigma$

$\therefore \mathrm{Det}(-\Omega) = -\sigma k \quad \left[\because \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc \right]$

$\therefore \mathrm{Det}(-\Omega) = \begin{bmatrix} p & -k \\ -\sigma & 0 \end{bmatrix}$

$= p.0 - (-k.-\sigma)$

$= -\sigma k < 0.$

if $A = \begin{bmatrix} 2 & 1 & 5 \\ 2 & 3 & 4 \\ 0 & 1 & 0 \end{bmatrix}$

$\therefore \mathrm{Tr}(\Omega) = A_{11} + A_{22} + A_{33}.$

$= 2 + 3 + 0$

$= 5.$

Here, $\mathrm{Tr}(-\Omega) = p + 0$

$= p > 0.$

$x_t = E_t x_{t+1} + \Delta + \sigma p - \sigma(i_t - E_t \pi_{t+1}).$

$= E_t x_{t+1} + \sigma\left(\frac{\Delta}{\sigma} + p\right) - \sigma(i_t - E_t \pi_{t+1}).$

$= E_t x_{t+1} + \sigma r^n - \sigma(i_t - E_t \pi_{t+1}).$

$\left[\because r^n = \frac{\Delta}{\sigma} + p\right]$

$= E_t x_{t+1} - \sigma(i_t - E_t \pi_{t+1} - r^n).$